

Markov Blanket Optimization with Graph Diffusion

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Preliminary

Markov Blanket Optimization

The problem of high correlation between parents of a variable. It is difficult to tell the causality when one observes high dependency between the parents.

The complexity of the model selection. The feature selection complexity is $C = 2^n - 1$. The causal model selection complexity is the number of the selected edges given the number of nodes. For n nodes the number of the edges is $m = 2(n^2 - n)$.

Include the complexity of the Markov blanket approach

$$C_{\text{MB}} =$$

The goal is to select a proper number of edges. It must be DAG, one-connected, minimal complexity.

Put the SCM definition here

Minimax problem statement

Minimum complexity of union and maximum probability of elements of the Markov blanket.

The elements of the MB are

1. probability of the vertex prior post

2. probability of the edge prior post
- 3.
- 4.
- 5.